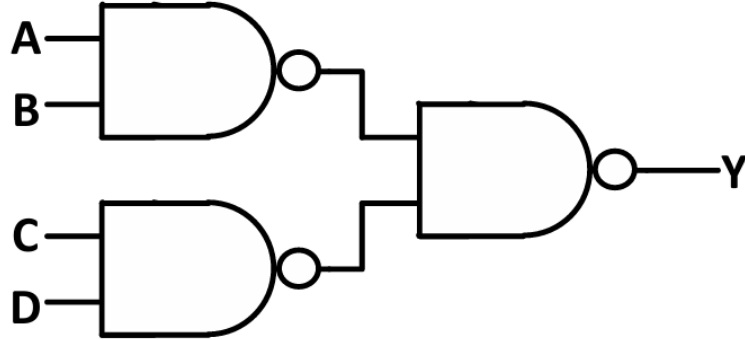


The objective of this experiment is to analyze a combinational logic circuit, simplify the obtained Boolean expression, and verify the output through hardware implementation using Arduino.



$$Y = AB + CD \quad (1)$$

1 Hardware Requirements

1. Arduino Uno
2. Common Anode Seven Segment Display
3. Breadboard
4. Jumper Wires
5. 220 Ω Resistors
6. Android Device with Termux and PlatformIO

2 Circuit Connections

Segment	Arduino Pin
a	D2
b	D3
c	D4
d	D5
e	D6
f	D7
g	D8
COM	+5V

Table 1: Seven Segment Connections

The common anode terminals of the display are connected to +5V.

3 Implementation

The Boolean expression

$$[Y = AB + CD] \quad (2)$$

was implemented in Arduino using predefined input combinations.

The output obtained from the logic expression was displayed on the Common Anode Seven Segment Display.

Two test cases were considered:

- Input Set 1: $A = 1$,
 $B = 1$,
 $C = 0$,
 $D = 0$
- Input Set 2: $A = 0$,
 $B = 1$,
 $C = 0$,
 $D = 1$

For Input Set 1,

$$Y = (1)(1) + (0)(0) = 1 \quad (3)$$

The display shows digit 1.

For Input Set 2,

$$Y = (0)(1) + (0)(1) = 0 \quad (4)$$

The display shows digit 0.

4 Observation

The output displayed on the seven segment display matched the result obtained from the Boolean expression.

A	B	C	D	Y=AB+CD
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Table 2: Truth Table for $Y = AB + CD$

5 Result

The Boolean function

$$Y = AB + CD \quad (5)$$

was successfully implemented using Arduino Uno and a Common Anode Seven Segment Display.

